

ECON 108

INTRODUCTION TO ECONOMICS

INSTRUCTOR: ALEJANDRO MELO PONCE

Demand and supply

Outline for today...

Demand and the Law of Demand

The Demand Schedule

Shifts in Demand

Supply and the Law of Supply

The Supply Schedule

Shifts in Supply

Putting Demand and Supply Together

Markets

Markets:

- All of the arrangements that individuals have for exchanging with one another
- One of the most important activities in these markets is the determination of prices
- A market need not be physical location

Markets

Example of Markets:

- Markets for gasoline
- Markets for labor
- Stock market
- Markets for Super Bowl tickets
- Markets for organ donors
- Electronic marketplaces

Relative Prices and Money Prices

Money Price:

- The price of a good or service expressed in today's dollars
- Also called ***absolute*** or ***nominal*** price

Relative Prices and Money Prices

Relative Price:

- The money price of one commodity divided by the money price of another commodity
- It represents the number of units of one commodity that must be sacrificed to purchase one unit of another commodity

Relative Prices and Money Prices

Example:

- Nominal price of apples: \$0.30 each
- Nominal price of bananas: \$0.19 each
- Relative price of apples: $0.30/0.19$, one apple buys approx. 1.57 bananas
- Relative price of bananas: $0.19/0.30$, one banana buys approx. 0.633 apples

DEMAND AND THE LAW OF DEMAND

Demand

A schedule showing how much of a good or service people will purchase at any price during a specific time period, other things being constant.

Can be interpreted *horizontally* or *vertically*

DEMAND AND THE LAW OF DEMAND

Law of Demand

The observation that there is a negative, or inverse, relationship between the price of any good or service and the quantity demanded, other things constant.

DEMAND AND THE LAW OF DEMAND

Law of Demand

- When the price of a good goes up, people buy less of it, other things being equal
- When the price of a good goes down, people buy more of it, other things being equal

THE DEMAND SCHEDULE

Demand Schedule

- The demand schedule is a table relating prices to quantity demanded
- We must consider:
 - The time dimension
 - Constant-quality units
 - Spatial dimension

THE DEMAND SCHEDULE

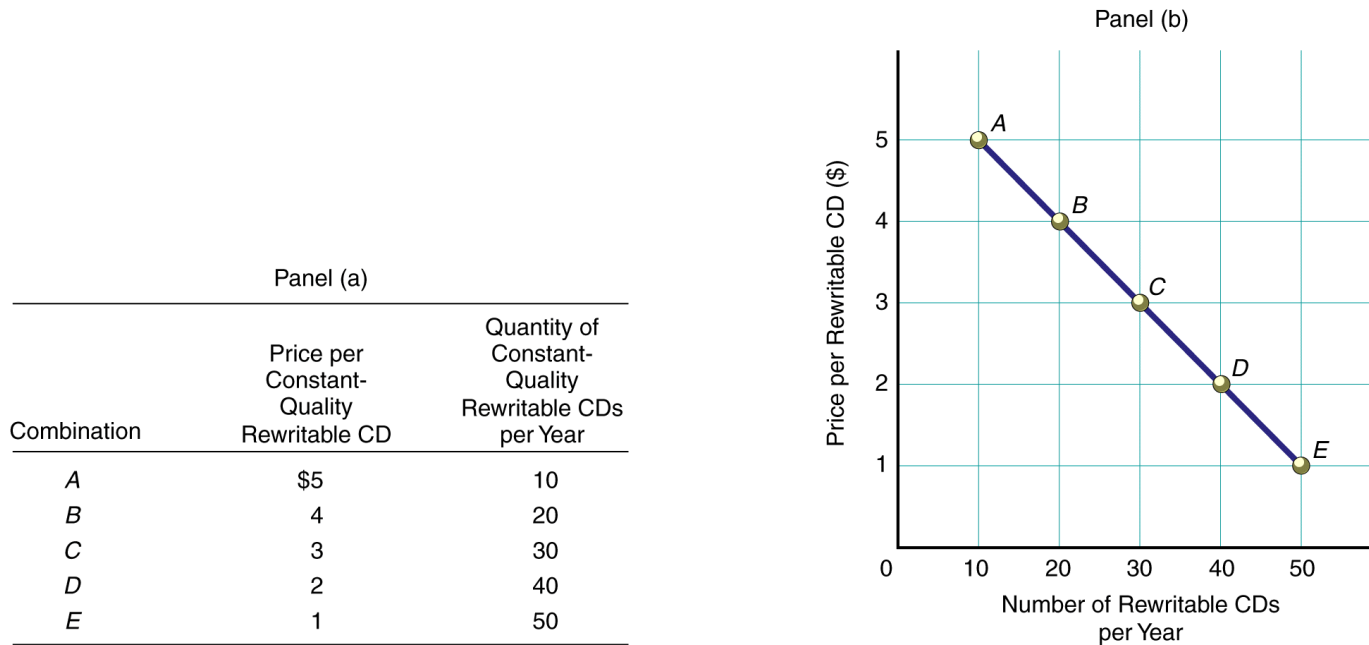


Figure 3-1

THE DEMAND SCHEDULE

Demand Curve

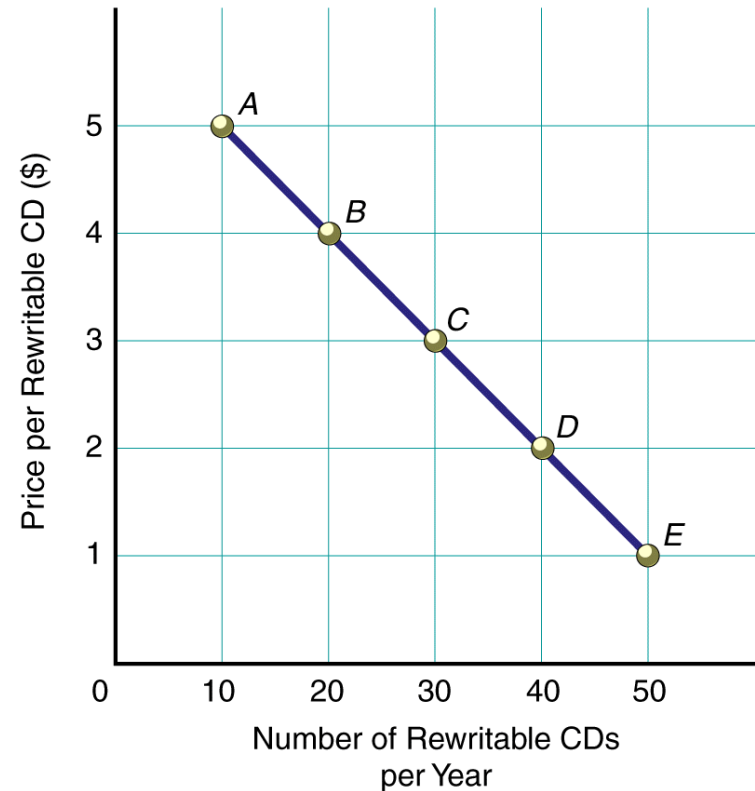
- Is a graphical representation of the demand schedule
- A negatively sloped curve showing the inverse relationship between the price and the quantity demanded when all other things are constant

THE DEMAND SCHEDULE

Panel (a)

Combination	Price per Constant-Quality Rewritable CD	Quantity of Constant-Quality Rewritable CDs per Year
<i>A</i>	\$5	10
<i>B</i>	4	20
<i>C</i>	3	30
<i>D</i>	2	40
<i>E</i>	1	50

Panel (b)



THE DEMAND SCHEDULE

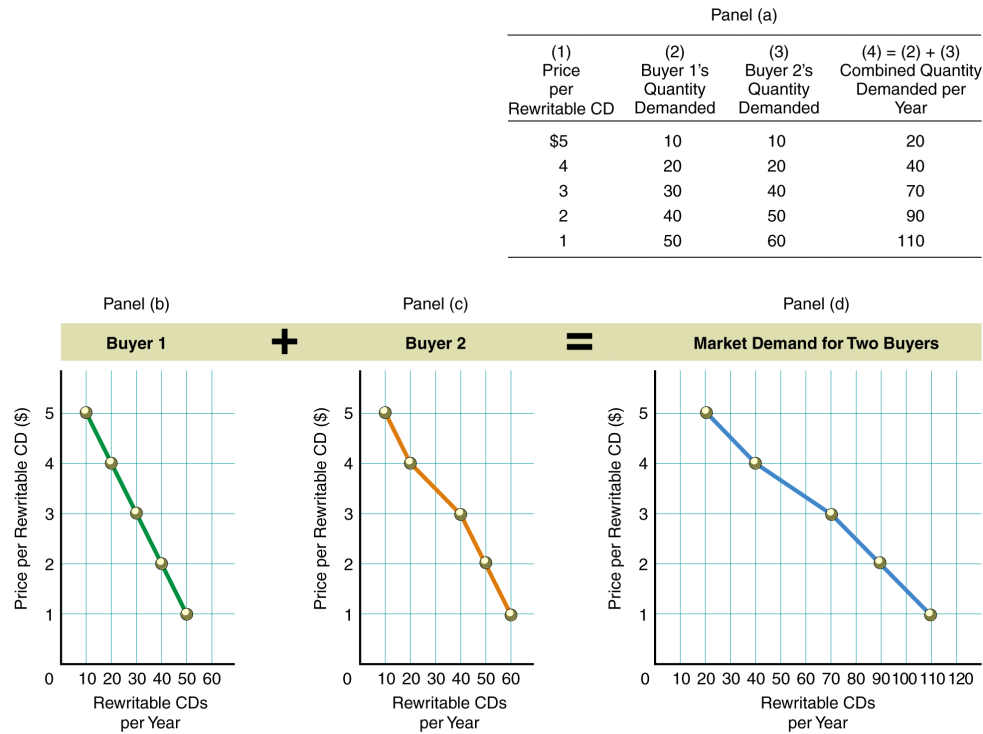


Figure 3-2

THE DEMAND SCHEDULE

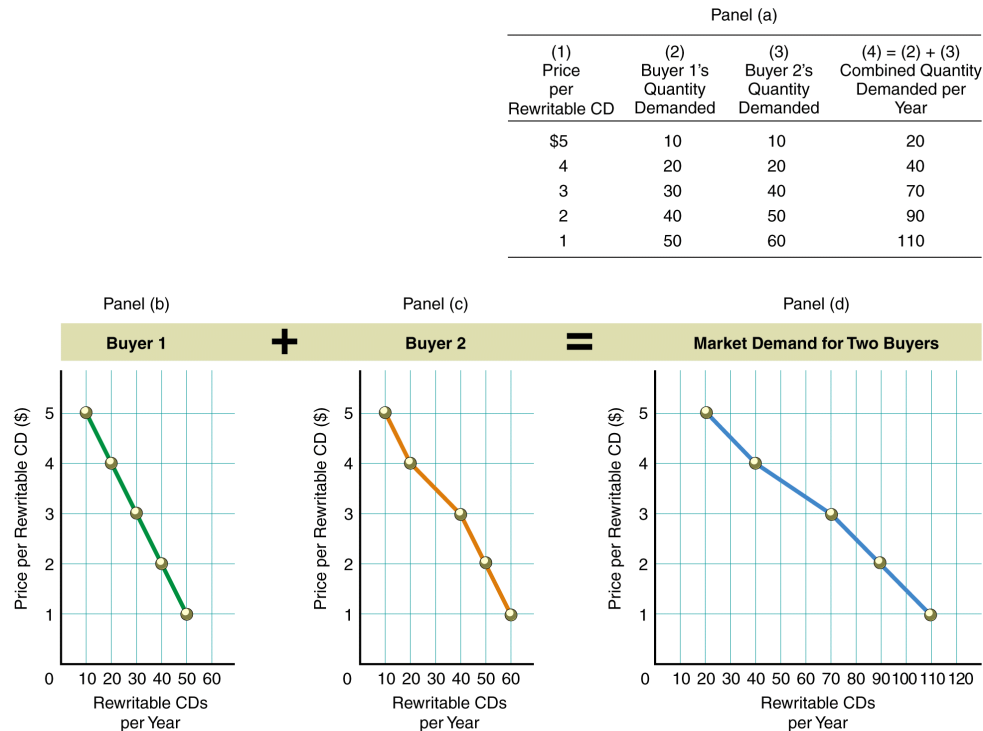


Figure 3-2

THE DEMAND SCHEDULE

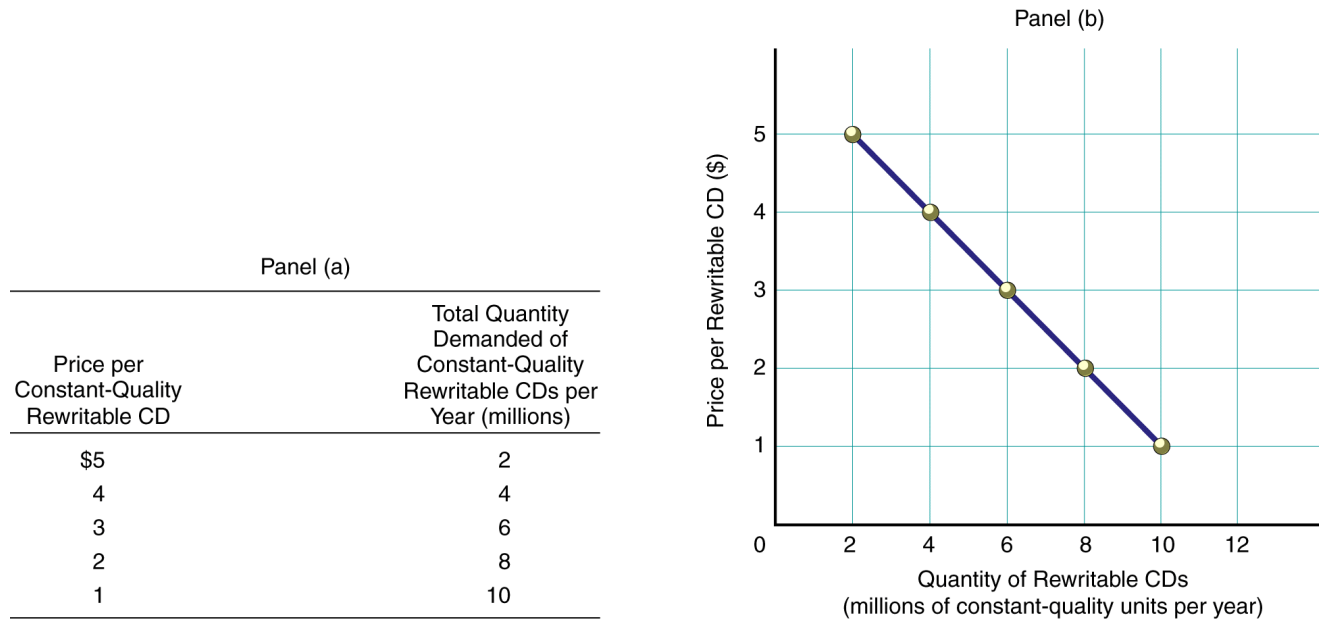


Figure 3-3

SHIFTS IN DEMAND

Other Determinants of Demand

- Income
- Tastes and Preferences
- Price of Related Goods (substitutes and complements)
- Expectations (future prices, future income, future availability of the commodity)
- Population Growth

SHIFTS IN DEMAND

We MUST be able to distinguish between a change in the quantity demanded and a change in demand.

Change in quantity demanded

A change in quantity demanded is a movement along the demand curve

- A change in quantity demanded occurs when the price of the commodity changes, all other things equal
- This is a movement along the curve and **NOT** a shift of the curve.

Change in demand

A change or a shift in demand is a movement of the entire curve.

- A change in demand occurs when one of the elements underlying the demand curve shifts (income, price of related goods, tastes, etc.)
- This IS a shift in the curve.

Be ready for the assignments...

Suppose that short skirts that were fashionable in the 1990s become unfashionable in the late 2000s. If other factors were held constant, then there would be

- a. a rightward movement along the supply curve.
- b. a rightward shift of the demand curve.
- c. a leftward shift in the demand curve.
- d. a leftward movement along the supply curve.

Be ready for the assignments...

If the price of airline travel in Europe falls and the demand for train travel in Europe also falls, then the two goods are

- a.complements.
- b.normal goods.
- c.substitutes.
- d.inferior goods.

Be ready for the assignments...

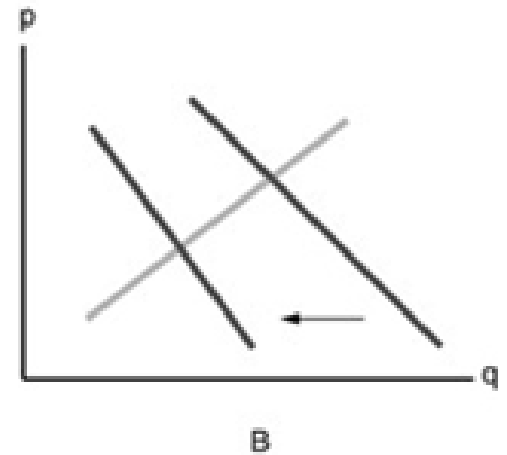
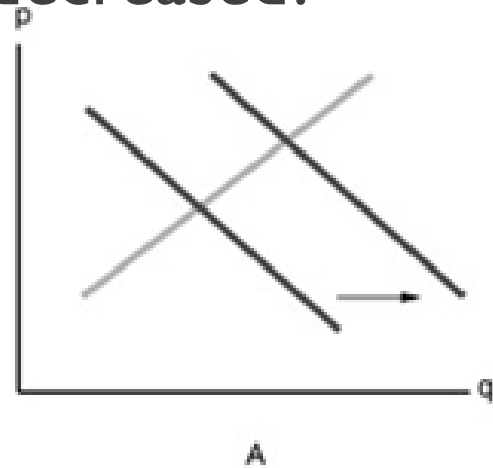
In economic terminology, a normal good is a good

- a. on which a monetary value cannot be placed.
- b. that is liked only by normal people.
- c. for which demand increases when price increases.
- d. for which demand increases when income increases.

Be ready for the assignments...

Based on the figure below, which diagram depicts the effect on the demand for Cheerios when the price of Milk has decreased?

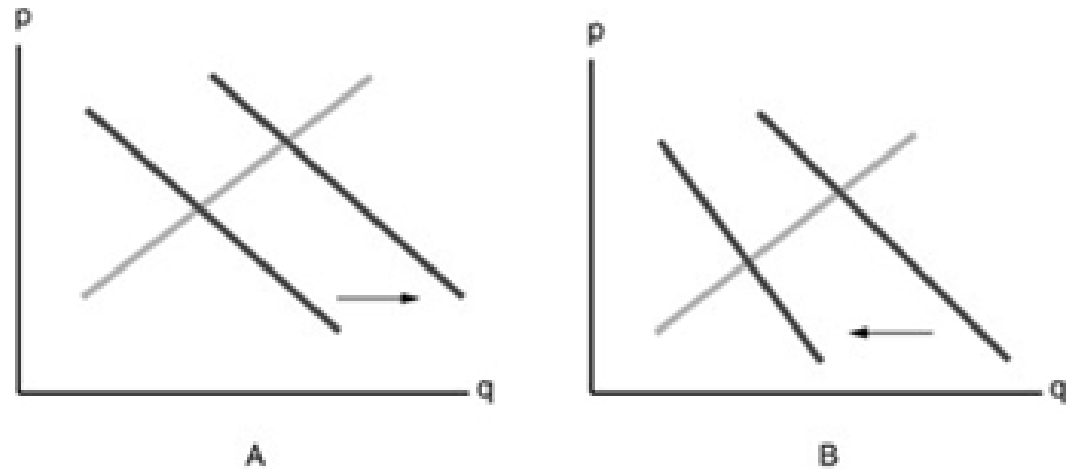
- a. A
- b. B
- c. Neither graph
- d. Both graphs



Be ready for the assignments...

Based on the figure below, which diagram depicts the effect on the demand for Cheerios when the price of Apple Jacks cereal has decreased?

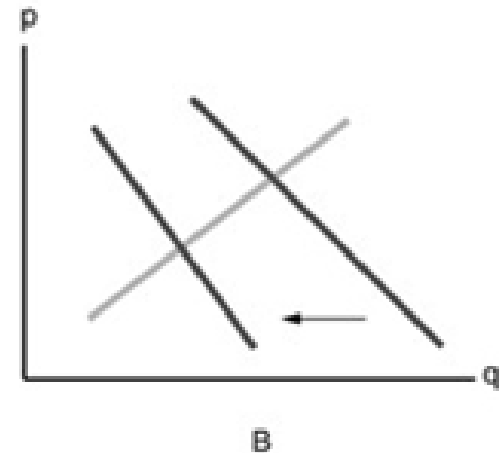
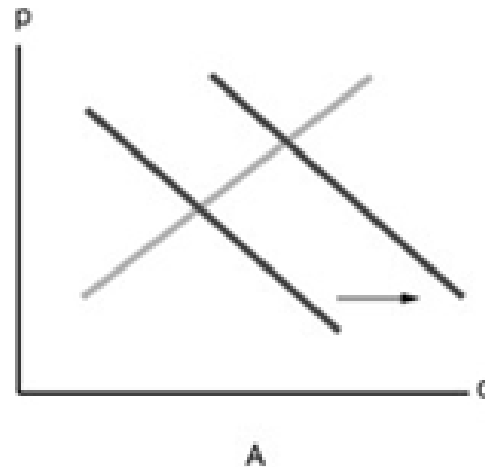
- a. A
- b. B
- c. Neither graph
- d. Both graphs



Be ready for the assignments...

Based on the above figure, which diagram depicts the effect on the demand for Cheerios when income has decreased and cereal is inferior?

- a. A
- b. B
- c. Neither graph
- d. Both graphs



Be ready for the assignments...

Which of the following illustrates the law of demand?

1. More people watch college basketball in March than in November.
2. The number of long distance calls in the United States is greater on Christmas than on Valentine's Day.
3. College enrollment increases when federal tuition grants are readily available to students.
4. The prevailing wage rate in an industry determines how many people choose to work in the industry.

SUPPLY AND THE LAW OF SUPPLY

Supply

A schedule showing the relationship between price and quantity supplied for a specific period of time, other things being constant.

Can be interpreted *horizontally* or *vertically*

SUPPLY AND THE LAW OF SUPPLY

Law of Supply

The observation that there is a positive, or direct relationship between the price of any good or service and the quantity supplied, other things constant.

SUPPLY AND THE LAW OF SUPPLY

Law of Supply

- When the price of a good increases, the quantity supplied of the good increases, other things being equal.
- When the price of a good decreases, the quantity supplied of the good decreases, other things being equal.

THE SUPPLY SCHEDULE

Supply Schedule

- The supply schedule is a table relating prices to quantity supplied.
- We must consider:
 - The time dimension
 - Constant-quality units
 - The spatial dimension

THE SUPPLY SCHEDULE

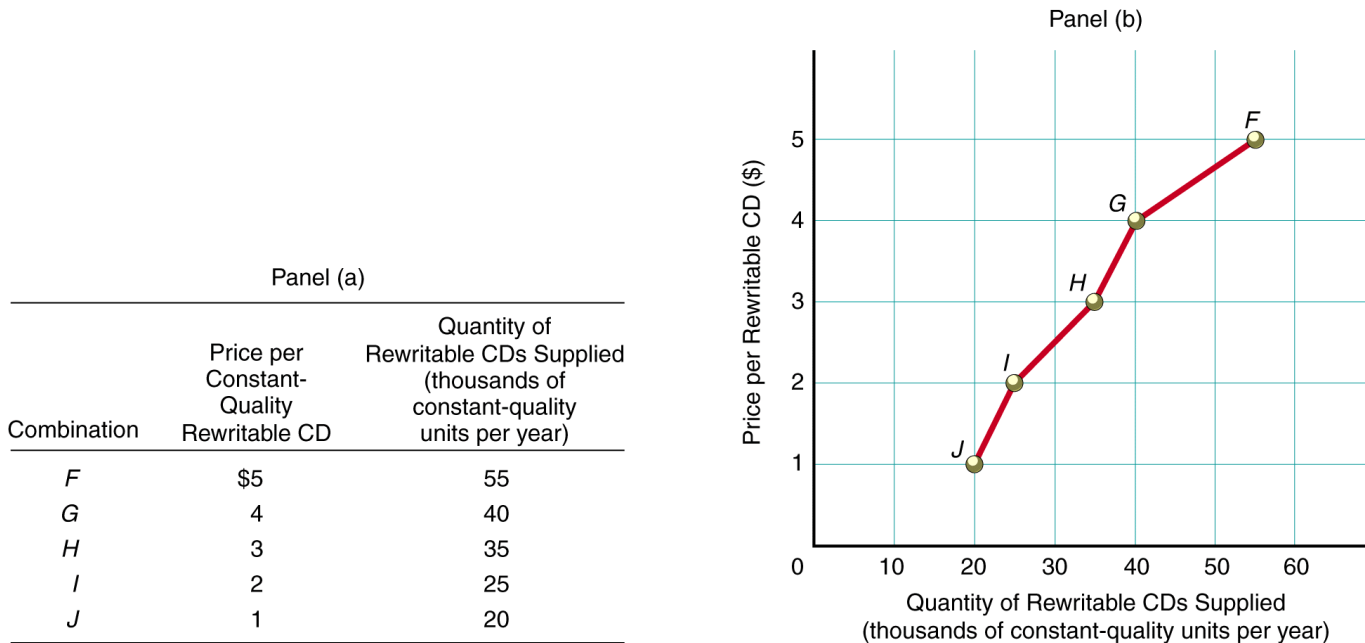


Figure 3-6

THE SUPPLY SCHEDULE

Supply Curve

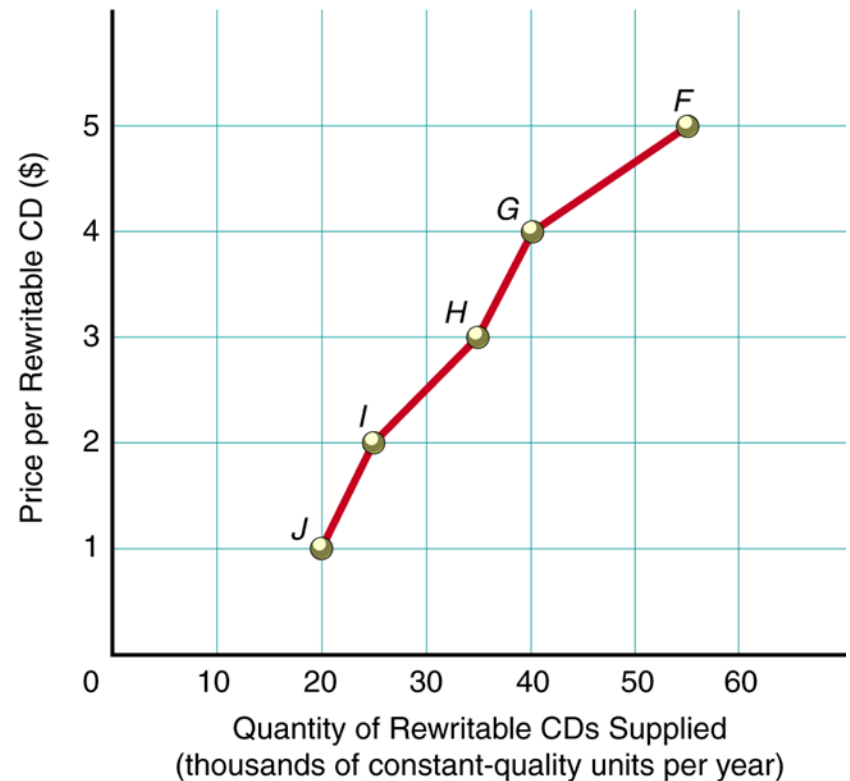
- Is a graphical representation of the supply schedule
- A positively sloped curve showing the direct relationship between the price and the quantity supplied when all other things are constant

THE SUPPLY SCHEDULE

Panel (a)

Combination	Price per Constant-Quality Rewritable CD	Quantity of Rewritable CDs Supplied (thousands of constant-quality units per year)
<i>F</i>	\$5	55
<i>G</i>	4	40
<i>H</i>	3	35
<i>I</i>	2	25
<i>J</i>	1	20

Panel (b)



THE SUPPLY SCHEDULE

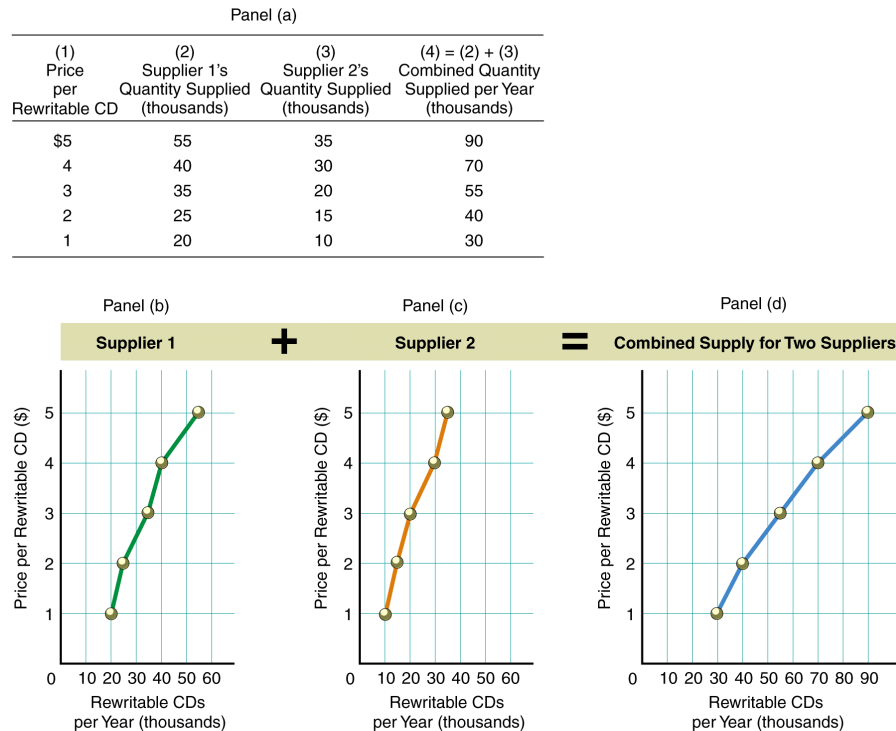
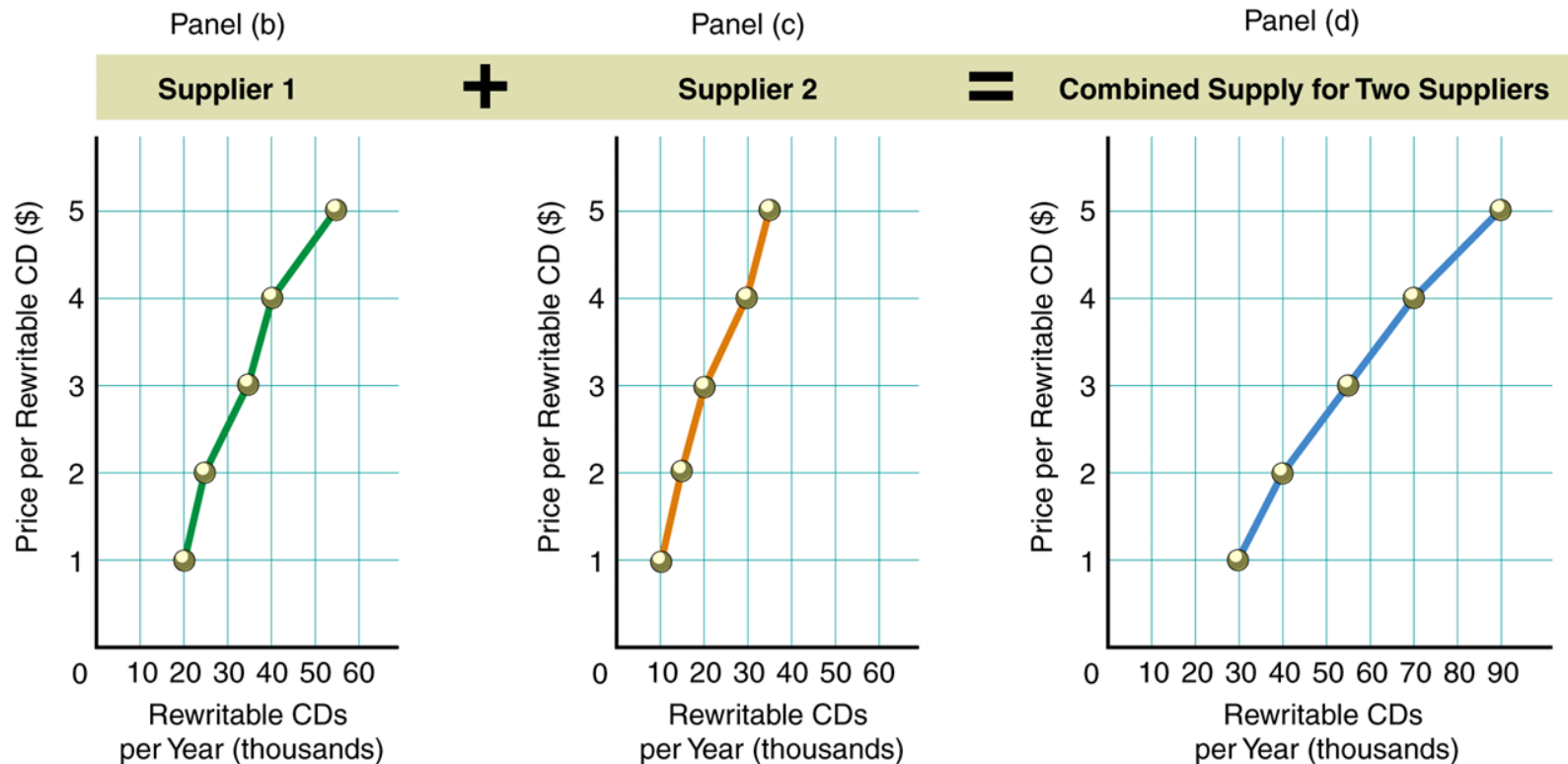


Figure 3-7

THE SUPPLY SCHEDULE



SHIFTS IN SUPPLY

Other Determinants of Supply

- Cost of inputs
- Technology and productivity
- Taxes and subsidies
- Expectations (future prices)
- Number of firms in industry
- Weather

SHIFTS IN SUPPLY

We MUST be able to distinguish between a change in the quantity supplied and a change in supply.

Change in quantity supplied

A change in quantity supplied is a movement along the supply curve

- A change in quantity supplied occurs when the price of the commodity changes, all other things equal
- This is a movement along the curve and **NOT** a shift of the curve.

Change in supply

A change or a shift in supply is a movement of the entire curve.

- A change in supply occurs when one of the elements underlying the supply curve changes (cost of inputs, technology and productivity, etc.)

Be ready for the assignments...

A supply schedule

- a. can be used to generate a supply curve.
- b. is a table reflecting the inverse relationship between price and quantity supplied.
- c. shows what happens to quantity supplied when price is held constant.
- d. all of the above.

Be ready for the assignments...

Suppose that because of unseasonably cold weather in Florida, a significant portion of the orange crop has been lost to freezing temperatures. This statement means that

- a. the demand for oranges will rise.
- b. the quantity supplied of oranges will fall.
- c. the amount of oranges available at various prices will decline.
- d. the price of oranges will fall.

Be ready for the assignments...

Sugar is an input used to produce cereal. Suppose that the price of sugar rises. As a result,

- a. the supply curve for sugar will shift to the right.
- b. the supply curve for sugar will shift to the left.
- c. the supply curve for cereal will shift to the right.
- d. the supply curve for cereal will shift to the left.

Be ready for the assignments...

Which of the following will shift today's supply curve to the right?

- a. Input prices rise.
- b. Sales taxes increase.
- c. Prices are expected to be higher in the future.
- d. Prices are expected to be lower in the future.

PUTTING DEMAND AND SUPPLY TOGETHER

Panel (a)

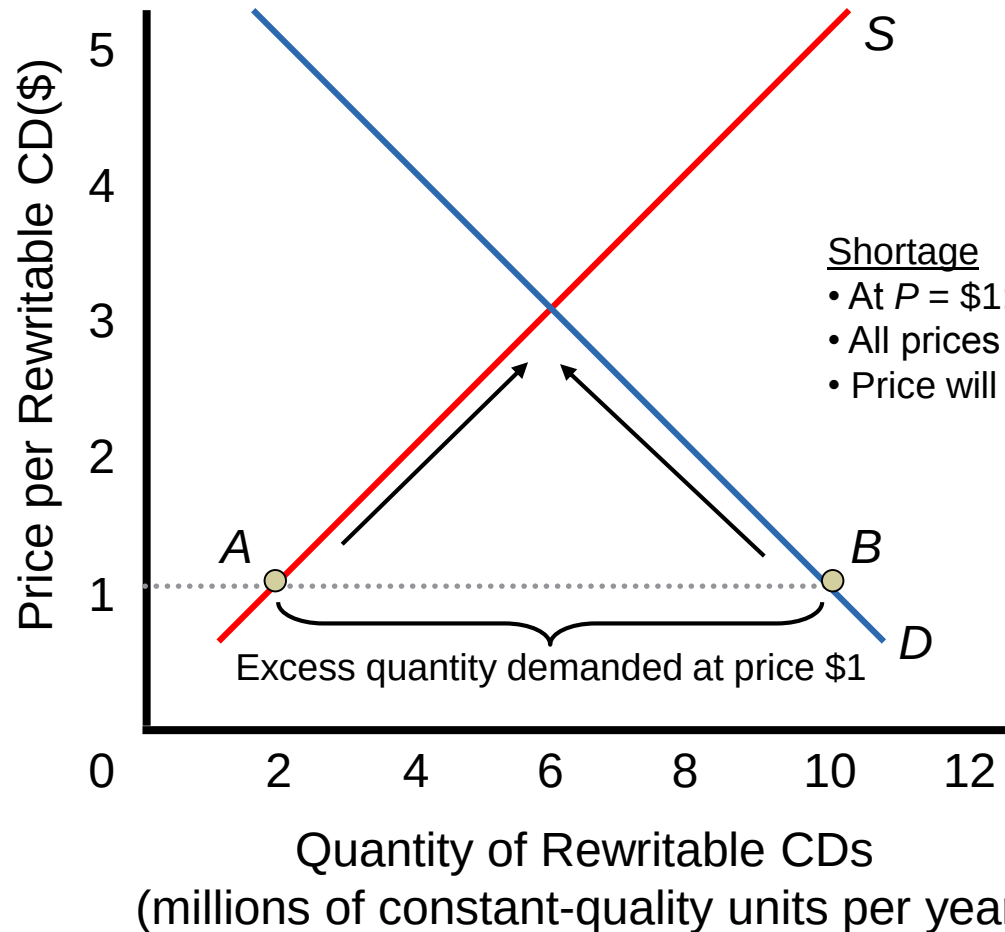
(1)	(2)	(3)	(4)	(5)
Price per Constant-Quality Rewritable CD	Quantity Supplied (rewritable CDs per year)	Quantity Demanded (rewritable CDs per year)	Difference (2) – (3) (rewritable CDs per year)	Condition
\$5	10 million	2 million	8 million	Excess quantity supplied (surplus)
4	8 million	4 million	4 million	Excess quantity supplied (surplus)
3	6 million	6 million	0	Market clearing price—equilibrium (no surplus, no shortage)
2	4 million	8 million	–4 million	Excess quantity demanded (shortage)
1	2 million	10 million	–8 million	Excess quantity demanded (shortage)

PUTTING DEMAND AND SUPPLY TOGETHER

Market Clearing Price

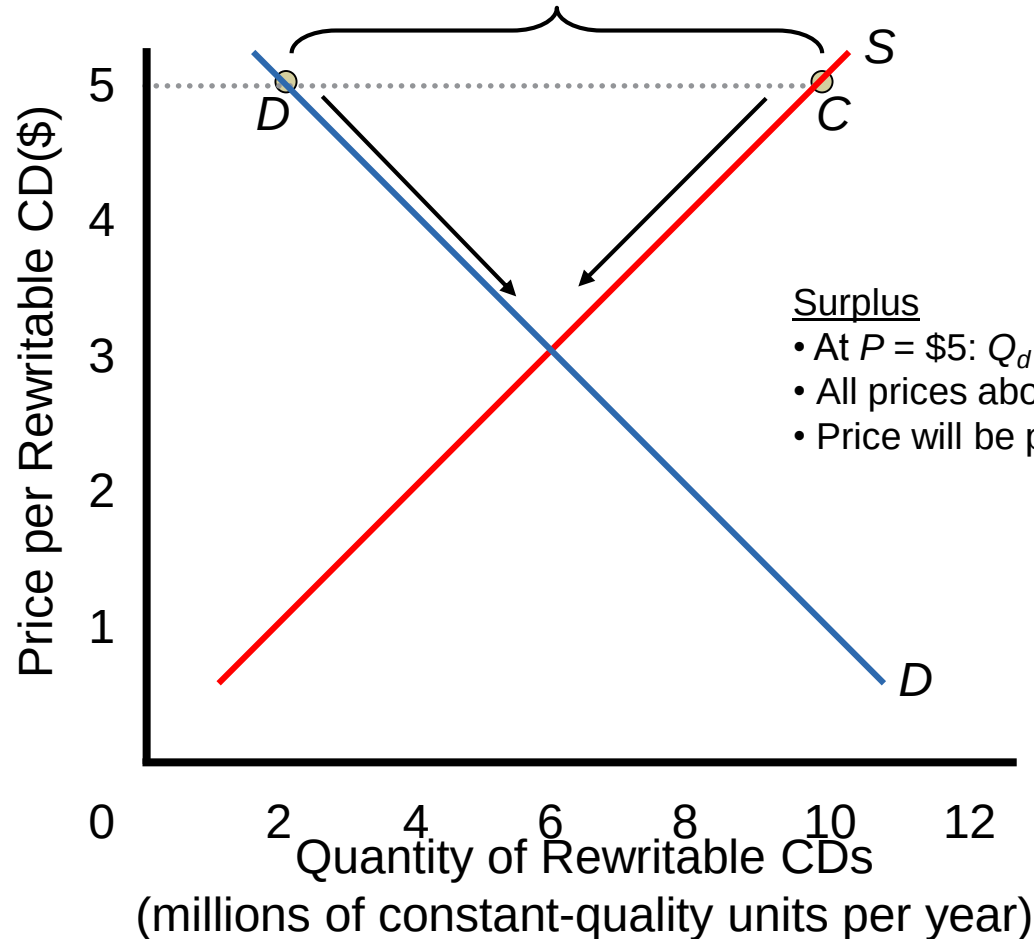
The market clearing price or equilibrium price is the price at which quantity demanded equals quantity supplied; the price where the demand curve intersects the supply curve

PUTTING DEMAND AND SUPPLY TOGETHER



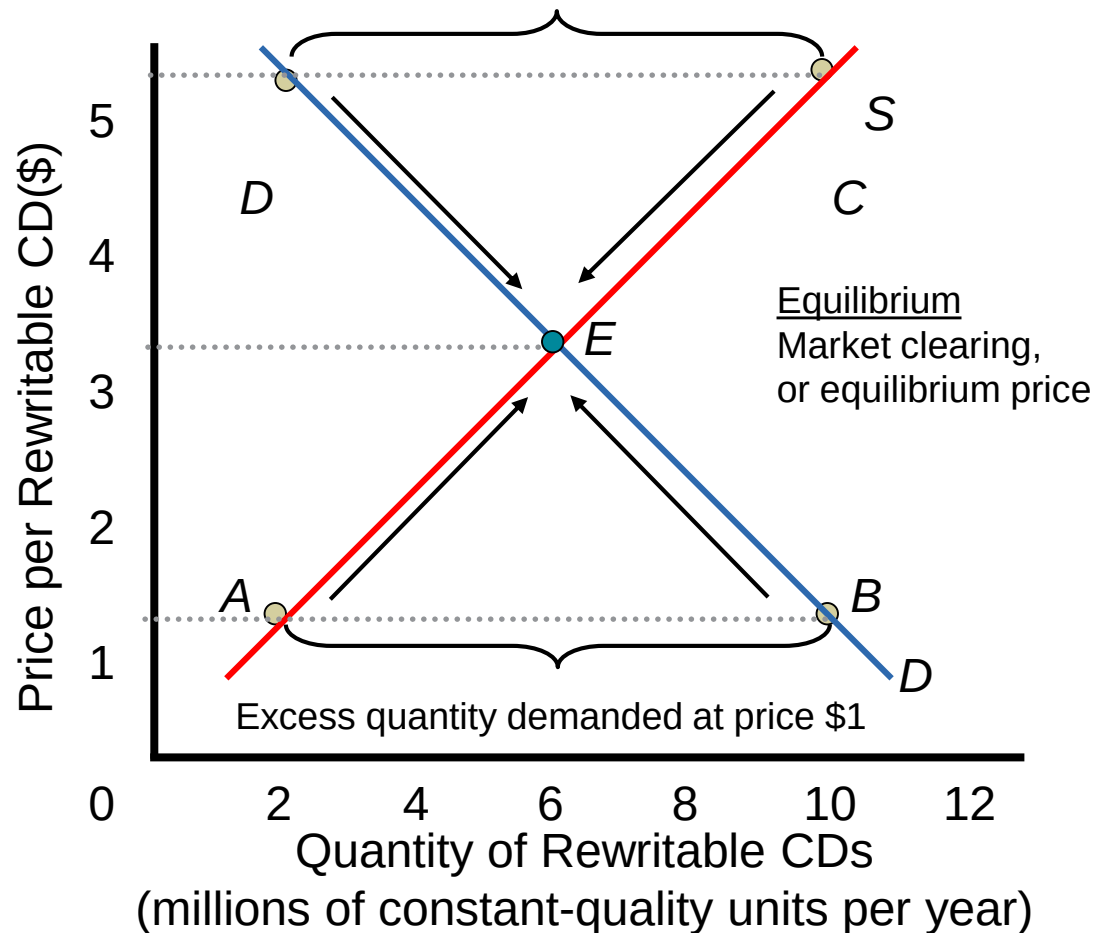
PUTTING DEMAND AND SUPPLY TOGETHER

Excess quantity demanded at price \$5



PUTTING DEMAND AND SUPPLY TOGETHER

Excess quantity demanded at price \$5



PUTTING DEMAND AND SUPPLY TOGETHER

Equilibrium

- The situation when quantity supplied equals quantity demanded at a particular price

PUTTING DEMAND AND SUPPLY TOGETHER

Excess Demand (Shortages)

- The situation when quantity demanded is greater than quantity supplied
- Is not the same as scarcity

PUTTING DEMAND AND SUPPLY TOGETHER

Excess Supply (Surpluses)

- The situation when quantity supplied is greater than quantity demanded

Be ready for the assignments...

The following equation represents the market demand for Ipads

$$Q_d = 1600 - 2P.$$

- a. Represent the above demand in a graph.
- b. Find the quantity demanded at a price of \$200. Show this quantity in the graph you drew for part a. Repeat for a price of \$300.

Be ready for the assignments...

The following equation represents the market supply of pizzas

$$Q_s = 2p$$

- a. Represent the above supply in a graph.
- b. Find the quantity supplied at a price of \$10. Show this quantity in the graph you drew for part a. Repeat for a price of \$20.

Be ready for the assignments...

The following equations describe the market for car rentals in Long Island,

$$Q_s = 2P \text{ and } Q_d = 80 - 2P.$$

- a. Find the equilibrium price and quantity.
- b. Draw and clearly label both supply and demand.
- c. Explain what would happen in the market if the price was \$10. Repeat if the price is \$25.